MILITARY SPECIFICATION

CLOTH, KNITTED, FLAME RESISTANT RAYON/WOOL PILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a sliver knit pile fabric. The pile is 65 percent flame resistant rayon, and 35 percent wool, while the backing is 100 percent para-aramid.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging of Woolen, Worsted and Wool Blend (Synthetic Fiber, Cotton) Fabrics

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1491 - Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

RULES AND REGULATION UNDER THE WOOL PRODUCTS LABELING ACT OF 1939

(Copies may be obtained from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8 Colorfastness to Crocking. AATCC Crockmeter Method

(Copies may be obtained from AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D-2130-78 Diameter of Wool and Other Animal Fibers by Microprojection.

(Copies may be obtained from American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). The standard sample is identified under piece number 87176.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Fiber.

3.3.1.1 Rayon. The rayon shall be permanent flame resistant rayon, when tested as specified in 4.6. It is manufactured by metering water-insoluble [poly (n-propoxyphosphazene)] into viscose solution from which rayon fibers are spun. The fibers contain the flame retardant which is highly dispersed in a matrix of cellulose.

3.3.1.2 Wool. The wool top shall be no lower in grade than 56's U.S. Standard, when tested as specified in 4.6. The use of reprocessed, reused or waste wool is prohibited.

3.3.1.3 Para-aramid. The para-aramid fiber shall not char at a temperature less than 675°F. The use of fiber other than the approved is prohibited.

3.3.2 Yarn.

3.3.2.1 Pile yarn. The pile yarn, which is in sliver form, shall be a staple blend of 65 (± 5) percent permanent flame resistant rayon by weight, and the remaining percentage wool, when tested as specified in 4.6.

3.3.2.2 Backing or ground yarn. The ground or backing yarn shall be 100 percent spun virgin para-aramid, when tested as specified in 4.6.

3.3.3 Knit. The fabric shall be a sliver knit, when tested as specified in 4.6. The pile side shall be designated as the face of the fabric. The back shall be a jersey knit (see 6.4).

3.4 Color. The color of the pile shall be Blue 3381 and shall match the standard sample. The color of the backing yarn shall be natural.

3.4.1 Matching. The finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial day light having a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

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3.4.2 Colorfastness. The finished cloth shall show colorfastness to laundering, wet dry cleaning, perspiration, and crocking equal to or better than the standard sample when tested as specified in 4.6. When no standard sample is available or when the standard sample is not referenced for colorfastness, the finished cloth shall show "fair" fastness to laundering, wet dry cleaning, and perspiration, and shall show a wet and dry crocking rating not lower than 4.

3.5 Finish. A flame resistant resin shall be applied to the back of the cloth as a bonding or anticurl agent (see 6.5), when tested as specified in 4.6. The cloth shall be evenly sheared to form a disoriented pile on the face equal to or better than the standard sample.

3.6 Physical requirements. The finished cloth shall conform to the physical requirements shown in Table I when tested as specified in 4.6.

TABLE I. - Physical requirements

Characteristics	Minimum	Maximum
Weight, oz/yd^2 (g/m^2)	10.5 (356)	12.5 (423.8)
Thickness (0.1 psi), inches (2.54 cm)	.150	.200
Wales per inch (2.54 cm)	14	-
Courses per inch (2.54 cm)	25	-
Burst strength, pounds (N)	150 (667.5)	-
Flame resistance,		
Initially and after 25 launderings:		
After flame time, seconds	-	3
After glow time, seconds	-	5
Char length, inches (2.54 cm)	-	5
Air permeability,		
ft^3/min/ft^2 (cm^3/sec/cm^2)	200 (101.6)	-

3.7 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width, inclusive of selvages when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe.

3.8 Dimensional stability. The finished cloth shall not shrink or elongate more than 3.0 percent in either the wale or course direction when tested as specified in 4.6.

3.9 pH. The pH value of the water extract of the finished cloth shall be no less than 3.3 nor more than 4.5 when tested as specified in 4.6.

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3.10 Pile height. The pile height of the finished cloth shall be 5/16 ± 1/32 inches when tested as specified in 4.6. The pile shall be vacuumed to remove the loose fibers.

3.11 Shearable pile percent. The total shearable pile percent, which is based on the finished fabric weight, shall be no more than 45 percent when tested as specified in 4.6.

3.12 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be put up on rolls, full width, face in with no tension, and shall be furnished in continuous lengths each not less than 30 yards (27.3m) in length as specified in PPP-P-1132. Splicing of the cloth is prohibited.

3.13 Wool content label. Each piece shall be labeled, or ticketed for wool content in accordance with the Wool Products Labeling Act of 1939.

3.14 Marking. The back side of the cloth shall be identified by stamping on that side with the word "Back" at each end of the roll.

3.15 Workmanship. The finished cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 3.2 and 6.2), the first article shall be visually examined for appearance, color and finish, and shall be tested for physical and chemical properties in accordance with the methods specified in 4.6.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined on one side only. The side examined, however, shall be alternated so that the alternate rolls are examined on the face and back respectively. All defects as defined in MIL-STD-1491 shall be classified and scored to the extent that they are clearly visible at normal inspection distance of 3 (± 1) feet on the side of the cloth under examination. Defects listed under 4.4.2.2, overall examination, shall not be scored in 4.4.2.1. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each wale-wise (lengthwise) yard (0.9m) or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard (0.9m). The lot size shall be expressed in units of one linear yard (0.9m) each. The sample size for this examination shall be based on inspection level II of MIL-STD-105, and the acceptance quality level (AQL) shall be 6.5 defects per 100 units (yards) (0.9m). The number of rolls from which the sample is to be selected shall be in accordance with Table II. An approximate equal number of yards shall be examined in each roll of the sample.

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4.4.2.2 Overall examination. Each defect listed below shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number (maximum number of defects acceptable) shall be as shown in Table II. The rolls selected for the yard-by-yard examination shall be included in this examination.

Defects
Edges cut, curled, folded, scalloped, or uneven, intermittently or throughout piece
Width less than specified
Color off shade, mottled, streaky or blotchy, shaded end to end, side to side, side to center, intermittently or throughout the piece
Pile character not as specified
Uneven shearing
Poor dye penetration
Overall uncleanness
Hand and finish not equal to standard sample
Objectionable odor
Face side of cloth not on inside

TABLE II. Sample size

Lot size in yards(.9m)	Sample size in rolls	Maximum number of defects acceptable
Up to 1200 1/	3	0
1201 to 3200	5	0
3201 to 10,000	8	0
10,001 to 35,000	13	0
35,001 to 150,000	20	1
150,001 and over	32	2

1/ If lot is fewer than three rolls, each roll in the lot shall be examined.

4.4.2.3 Examination for length.

4.4.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The sample size (number of rolls secected as sample) and the acceptance number (maximum number of defects acceptable) shall be as shown in table II. Any roll containing more than one piece or a splice shall also be considered a defect with respect to length.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

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4.4.2.4 Examination for marking and compliance with Wool Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample.

Back identification missing from either or both ends
Back identification on wrong side
Not labeled or ticketed in accordance with the rules and regulations under the Wool Products Labeling Act

4.5 Examination of packaging requirements. Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that preservation, packing, and marking complies with Section 5 requirements.

4.6 Tests. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods except for thickness. For thickness characteristics, the lot average of all sample units in the sample shall apply and the lot shall be unacceptable if the lot average fails to meet the minimum thickness requirements specified. The sample unit shall be 4 continuous yards (3.64m) full width of the finished cloth. The lot size shall be expressed in units of 1 yard (.9m) each. The lot shall be unacceptable, except for thickness if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)(0.9m)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. Test methods

Characteristic	Requirement Paragraph	Test Method
Fiber		
PFR Rayon identification	3.3.1.1	1/
Wool grade	3.3.1.2	4.6.1, ASTM D 2130-78 3/
Pile yarn blend	3.3.2.1	1/
Backing yarn identification	3.3.2.2	1/
Knit	3.3.3	1/
Colorfastness		
Laundering	3.4.2	5614
Wet dry cleaning	3.4.2	5622
Perspiration	3.4.2	5680
Crocking	3.4.2	AATCC-8
Resin finish	3.5	1/
Weight	3.6	5041
Thickness	3.6	5030
Wales/Courses per inch	3.6	5070
Burst strength	3.6	5120 2/
Flame resistance		
Initially	3.6	5903
After 25 launderings	3.6	5556,5903
Air permeability	3.6	5450
Dimensional stability	3.8	5556
pH	3.9	2811
Pile height	3.10	1/
Shearable pile percent	3.11	1/

1/ A certificate of compliance shall be submitted and shall be acceptable for this requirement.

2/ Tests shall be performed with pile facing down.

3/ One sample unit shall be taken from each lot and the lot shall be unacceptable if the sample fails to meet the required grade.

4.6.1 Strip wool. This method is intended for the stripping of dye from wool fibers in a wool or wool blend fabric to be used as the sample for wool grading in end items according to ASTM D 2130-78 entitled Diameter of Wool and Other Animal Fibers by Microprojection.

4.6.1.1 Test Specimen: Two specimens shall constitute a sample unit, each specimen shall consist of pile fibers weighing 5.0 (± 0.5) grams. Specimens shall be taken from areas at least two inches from the selvages and shall not be of the same wale and course yarns.

4.6.1.2 Apparatus: The equipment to be used are the launderometer and the canister described in Federal Test Method 5614, para. 4.1.1.

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Reagents

(1) Stripping chemicals The stripping chemicals shall be Zinc (Zn) (see 6.6) or Sodium (Na) sulfoxylated formaldehyde.

(2) Buffering agent: Acetic acid.

4.6.1.3 Procedure: Place one (1) gram of the stripping chemical in each of two canisters, add 400 ml of distilled water to each and adjust pH to 3.0 - 3.5 with acetic acid. Add the 5 gram fiber specimen to each canister. The canister shall be placed in the launderometer and rotated until the temperature reaches 208 ± 3°F. After the temperature of 208 ± 3°F has been reached, continue rotation for another 30 minutes. When the cycle has ended, remove the fiber, rinse with warm water (120°F) for 5 minutes and then cold water for 5 minutes. Air dry on a flat surface under ambient conditions.

5. PACKAGING

5.1 Packaging. Put-up and preservation packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A and Commercial. The cloth shall be put up and preserved in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use. The cloth covered by this document is intended for use in cold weather clothing.

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6.2 Ordering data. Procurement documents should specify the following.

a. Title, number and date of this document.
b. When a first article inspection is required (see 3.2 and 4.3), the item shall be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quality, and testing and approval of the first article.
c. Width, if other than specified (see 3.7).
d. Length, if other than specified (see 3.12).
e. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 Cloth, knitted, flame resistant rayon/wool pile, para-aramid back. Fabric conforming to style 221 of Glenoit Mills, Inc., New York, NY has been known to meet the requirements specified (see 3.3.3).

6.5 Bonding agent. The flame resistant acrylic/ethylene/vinyl chloride resin "SAGIBACK K-M4" of Astro Industries Inc., Morganton, NC is known to meet the requirements specified (see 3.5).

6.6 Parolite (see 4.6.1) can be purchased from:

Diamond Shamrock Chemicals Division
Process Chemical Division
350 Mt. Kemble Avenue
Morristown, NJ 07960-1931

6.7 Subject term (key word) listing.

a. Cloth, knitted, pile
b. Cloth, flame resistant
c. Fiber, flame resistant rayon/wool blend

Custodian:
Navy - NU
Air Force - 99

Preparing activity:
Navy - NU

Review Activity:
DPSC - CT
Air Force - 82

Project number: 8305-0154

User Interest
Navy - MC

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-24932	CLOTH, KNITTED, FLAME RESISTANT RAYON/WOOL PILE

3a NAME OF SUBMITTING ORGANIZATION

b ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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